

Inequality for square summable complex series

OPG-41335

L1 - TRACTABLE

PARTIALLY SOLVED

Analysis

ORIGINAL PROBLEM STATEMENT

Conjecture For all $\alpha = (\alpha_1, \alpha_2, \dots) \in l_2(\mathbb{C})$ the following inequality holds

$$\sum_{n \geq 1} |\alpha_n|^2 \geq \frac{6}{\pi^2} \sum_{k \geq 0} \left| \sum_{l \geq 0} \frac{1}{l+1} \alpha_{2^k(2l+1)} \right|^2$$